ENSO Intensity and California Statewide Precipitation: Water-Year Analysis

DATASCI 203 — Group 2

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1 Data

1.1 The ENSO series

The ONI file is pre-built: the December-February mean, NOAA's phase and strength classifications, 0/1 phase indicators, a one-year lag, and a lineartime trend. No derivation is needed here, which removes a whole class of possible error - but it also means the water-year alignment is inherited rather than verified, so the checks below matter.

Table 1: ENSO series validation.

Check	Value
Water years	76
Range	WY1950-WY2025
Mean ONI	0.01
SD ONI	1.03
Phase counts (La Nina / Neutral / El Nino)	29 / 20 / 27
Lag available for	75 years

1.2 The precipitation series

The preferred outcome is NOAA's statewide water-year total. If that file is notpresent, the chunk below falls back to a station-weighted composite of the regional wet-season data, and flags the substitution loudly. **The fallback is not the same construct** — it covers November through April, so it runs below the statewide annual figure (need to report).

Fallback outcome in use. The NOAA statewide series was not found, so results below use a station-weighted regional wet-season composite. Correlation with the ONI is 0.245 against 0.257 for the statewide series, so the pipeline is sound, but levels and every intercept are not comparable to published statewide figures.

1.3 Merge

All five models are estimated on these 75 water years. Restricting to rows with a defined lag costs the first sample year but makes AIC and the nested F-test legitimate comparisons rather than apples-to-oranges.

2 Descriptive analysis

Table 2: Descriptive statistics for the analysis variables.

Variable	N	Mean	SD	Min	Median	Max
Precipitation (in)	75	20.01	7.02	6.87	18.42	38.0
ONI (Dec-Feb, C)	75	0.03	1.03	-1.80	-0.10	2.6
Previous winter ONI (C)	75	0.02	1.04	-1.80	-0.10	2.6

Table 3: Precipitation by ENSO phase.

ENSO phase	Water years	Mean precip (in)	SD (in)	Driest	Wettest
La Niña	28	19.2	5.9	8.8	32.5
Neutral	20	18.6	6.5	10.4	32.5
El Niño	27	21.9	8.2	6.9	38.0

Table 4: Precipitation by ENSO strength bin. The very strong bin holds only a handful of years and cannot support inference on its own.

Strength	Water years	Mean precip (in)	SD (in)
neutral	20	18.6	6.5
weak	32	19.4	6.8
moderate	9	21.5	8.2
strong	11	20.2	5.2
very_strong	3	31.3	9.2

```
## Welch two-sample t-test, El Nino years vs all others: t = -1.66, df = 42.4, p = 0.105
```

2.1 Figures

3 Models

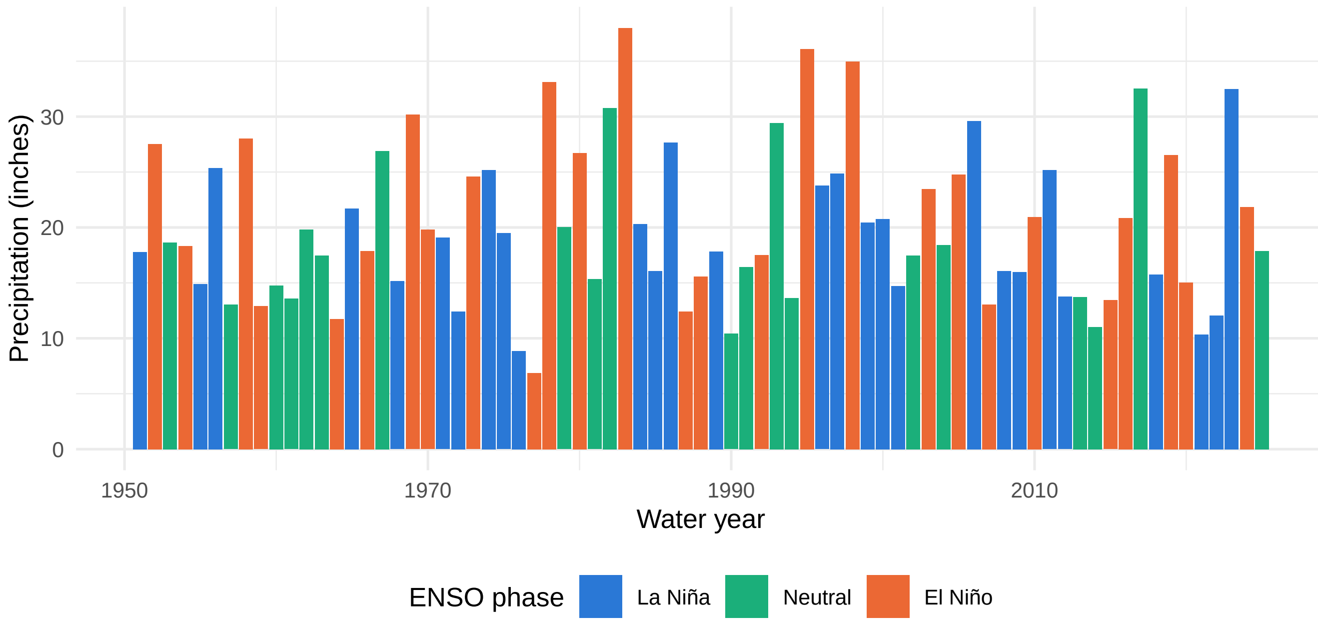

Figure 1: Statewide water-year precipitation, coloured by ENSO phase.

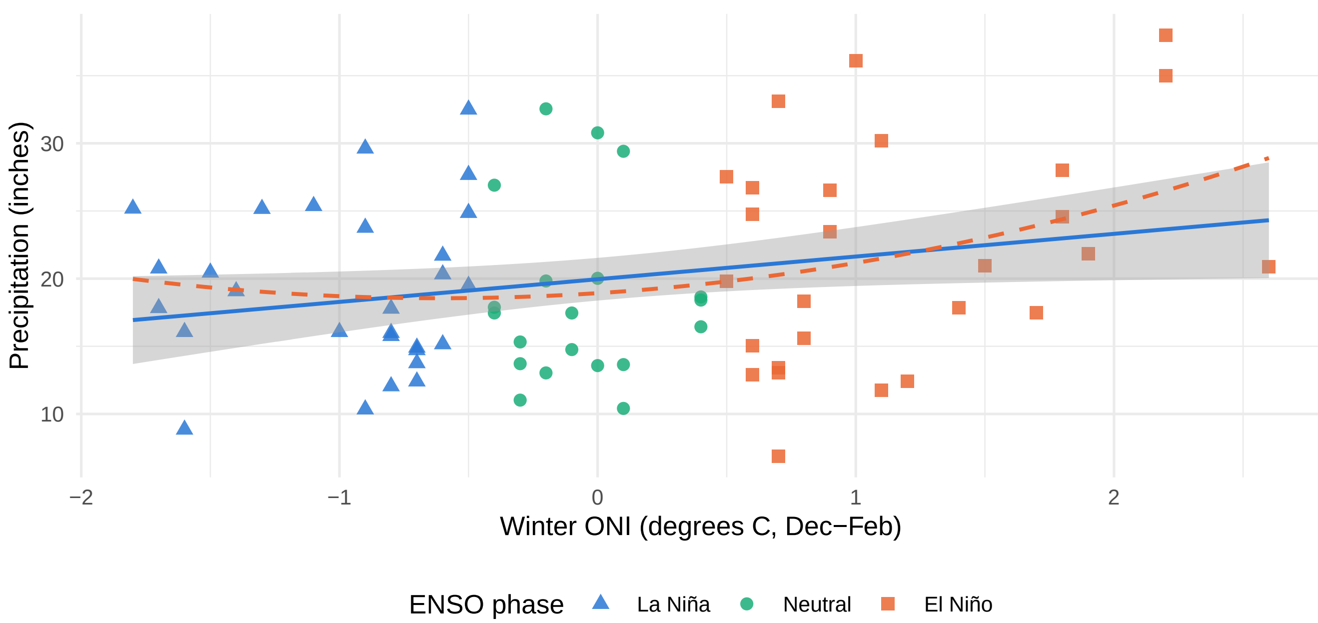

Figure 2: Precipitation against winter ONI, with linear and quadratic fits.

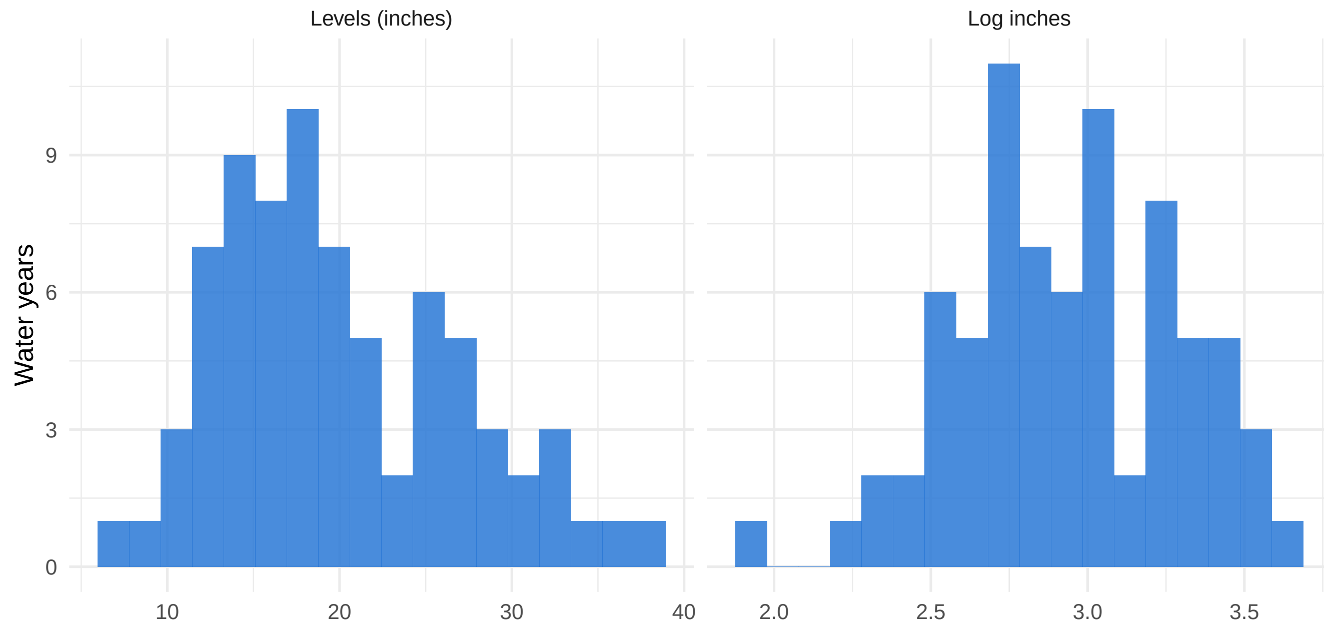

Figure 3: Precipitation is right-skewed; the log transform largely corrects it.

Model	Question
M1	Is there a linear ENSO-precipitation association?
M2	What does collapsing intensity into phase categories cost?
M3	Is the association confounded by last winter's ENSO or by trend?
M4	Is the relationship proportional rather than additive?
M5	Does the relationship steepen at extreme ONI values?

Table 5: ENSO intensity and California water-year precipitation.

	Dependent variable:				
		precip_in		log(precip_in)	precip_in
	M1	M2	M3	M4	M5
Winter ONI	1.677**		1.826**	0.072*	1.227
	(0.799)		(0.810)	(0.038)	(0.755)
Winter ONI squared		3.365			
		(2.138)			
El Nino		0.636			
		(1.823)			
La Nina			1.604**		
			(0.712)		
Previous winter ONI			0.004		
			(0.035)		
Time trend					1.005
					(0.650)
Constant	19.957***	18.564***	19.769***	2.933***	18.926***

Table 6: Effect sizes implied by Model 1. The final column is the one that matters for planning: it compares the ENSO signal to the year-to-year noise it must be detected against.

Scenario	Change vs mean (in)	As pct of mean	As pct of SD
One SD increase in winter ONI	1.72	8.60	24.50
Strong El Nino (ONI = +1.5)	2.52	12.57	35.82
Very strong El Nino (ONI = +2.3)	3.86	19.27	54.93
Strong La Nina (ONI = -1.5)	-2.52	-12.57	-35.82

3.2 Model comparison

Table 7: M4's outcome is on a different scale, so its R-squared and AIC are NOT comparable to the others. Compare it only to itself.

Model	R-squared	Adj R-squared	AIC
M1 linear	0.0600	0.0472	505.5321
M2 phase dummies	0.0438	0.0172	508.8184
M3 lag + trend	0.1163	0.0790	504.9001
M4 log outcome	0.0432	0.0301	59.6094
M5 quadratic	0.0924	0.0672	504.9025

```
## Analysis of Variance Table
##
## Model 1: precip_in ~ oni_djf
## Model 2: precip_in ~ oni_djf + oni_djf_lag1 + time_trend
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     73 3429.1
## 2     71 3223.7  2    205.37 2.2616 0.1116
```

```
## Analysis of Variance Table
##
## Model 1: precip_in ~ oni_djf
## Model 2: precip_in ~ oni_djf + I(oni_djf^2)
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     73 3429.1
## 2     72 3310.9  1    118.14 2.5691 0.1133
```

4 Regression assumptions

Table 8: Classical linear model assumption checks.

Assumption	Test	Value
IID / no serial correlation	Breusch-Godfrey(2)	LM = 1.04, p = 0.594
IID / no serial correlation	Lag-1 residual autocorrelation	-0.058
Homoskedasticity (levels)	Breusch-Pagan	p = 0.070
Homoskedasticity (log)	Breusch-Pagan	p = 0.509
Normal errors (levels)	Shapiro-Wilk	p = 0.025
Normal errors (log)	Shapiro-Wilk	p = 0.174
Skewness (levels)	-	0.59
Skewness (log)	-	-0.17

Table 9: Variance inflation factors, M3.

Term	VIF
oni_djf	1.01
oni_djf_lag1	1.01
time_trend	1.00

5 Reproducibility check

Table 10: Values computed independently in Python from the same two source files. The descriptive rows also match the published report exactly, which confirms the pipeline reproduces the original analysis. If the knit disagrees with any row here, something drifted.

Quantity	Expected
Descriptives n = 76 (WY1950-2025)	mean 22.18, SD 6.75, min 10.86, median 20.80, max 40.41
Correlation with ONI	0.257
Skewness levels -> log	0.65 -> 0.11
Phase groups (La Nina/Neutral/El Nino)	29 / 20 / 27
Welch t-test, El Nino vs rest	t = 1.77, p = 0.084
Model sample	n = 75 (WY1951-2025)
M1 slope	1.645 (HC1 SE 0.807, p = 0.042)
M1 intercept / R-squared	22.177 / 0.0620
M2 El Nino / La Nina	3.251 (p = 0.120) / 0.348 (p = 0.840)
M2 R-squared	0.0477 (below M1)
M3 ONI / lag / trend	1.765 (p = 0.028) / 1.379 (p = 0.047) / -0.027 (p = 0.425)
M4 semi-elasticity	0.062 (p = 0.061)
M5 quadratic term	1.011 (p = 0.122)

Nested F, M1 vs M3	2.03, p = 0.139
Nested F, M1 vs M5	2.81, p = 0.098
Breusch-Godfrey(2)	LM = 1.50, p = 0.471
Lag-1 residual autocorrelation	-0.098
Breusch-Pagan (levels / log)	p = 0.010 / p = 0.170
Shapiro-Wilk (levels / log)	p = 0.009 / p = 0.043

```
## R version 4.6.0 (2026-04-24)
## Platform: aarch64-apple-darwin24.6.0
## Running under: macOS Sequoia 15.5
##
## Matrix products: default
## BLAS:   /opt/homebrew/Cellar/openblas/0.3.33/lib/libopenblasp-r0.3.33.dylib
## LAPACK: /opt/homebrew/Cellar/r/4.6.0/lib/R/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] moments_0.14.1   kableExtra_1.4.1 knitr_1.51       car_3.1-5
##  [5] carData_3.0-6    stargazer_5.2.3  lmtest_0.9-40    zoo_1.8-15
##  [9] sandwich_3.1-1   lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0
## [13] dplyr_1.2.1      purrr_1.2.2      readr_2.2.0      tidyr_1.3.2
## [17] tibble_3.3.1     ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       xfun_0.57          lattice_0.22-9     tzdb_0.5.0
##  [5] vctrs_0.7.3        tools_4.6.0        generics_0.1.4     stats4_4.6.0
##  [9] parallel_4.6.0     pkgconfig_2.0.3    Matrix_1.7-5       RColorBrewer_1.1-3
## [13] S7_0.2.2           lifecycle_1.0.5    compiler_4.6.0     farver_2.1.2
## [17] textshaping_1.0.5  htmltools_0.5.9    yaml_2.3.12        Formula_1.2-5
## [21] pillar_1.11.1      crayon_1.5.3       abind_1.4-8        nlme_3.1-169
## [25] tidyselect_1.2.1   digest_0.6.39      stringi_1.8.7      labeling_0.4.3
## [29] splines_4.6.0      rprojroot_2.1.1    fastmap_1.2.0      grid_4.6.0
## [33] cli_3.6.6          magrittr_2.0.5     withr_3.0.2        scales_1.4.0
## [37] bit64_4.8.0        timechange_0.4.0   rmarkdown_2.31     bit_4.6.0
## [41] otel_0.2.0         hms_1.1.4          evaluate_1.0.5     viridisLite_0.4.3
## [45] mgcv_1.9-4         rlang_1.2.0        glue_1.8.1         xml2_1.5.2
## [49] svglite_2.2.2      rstudioapi_0.18.0  vroom_1.7.1        R6_2.6.1
## [53] systemfonts_1.3.2
```

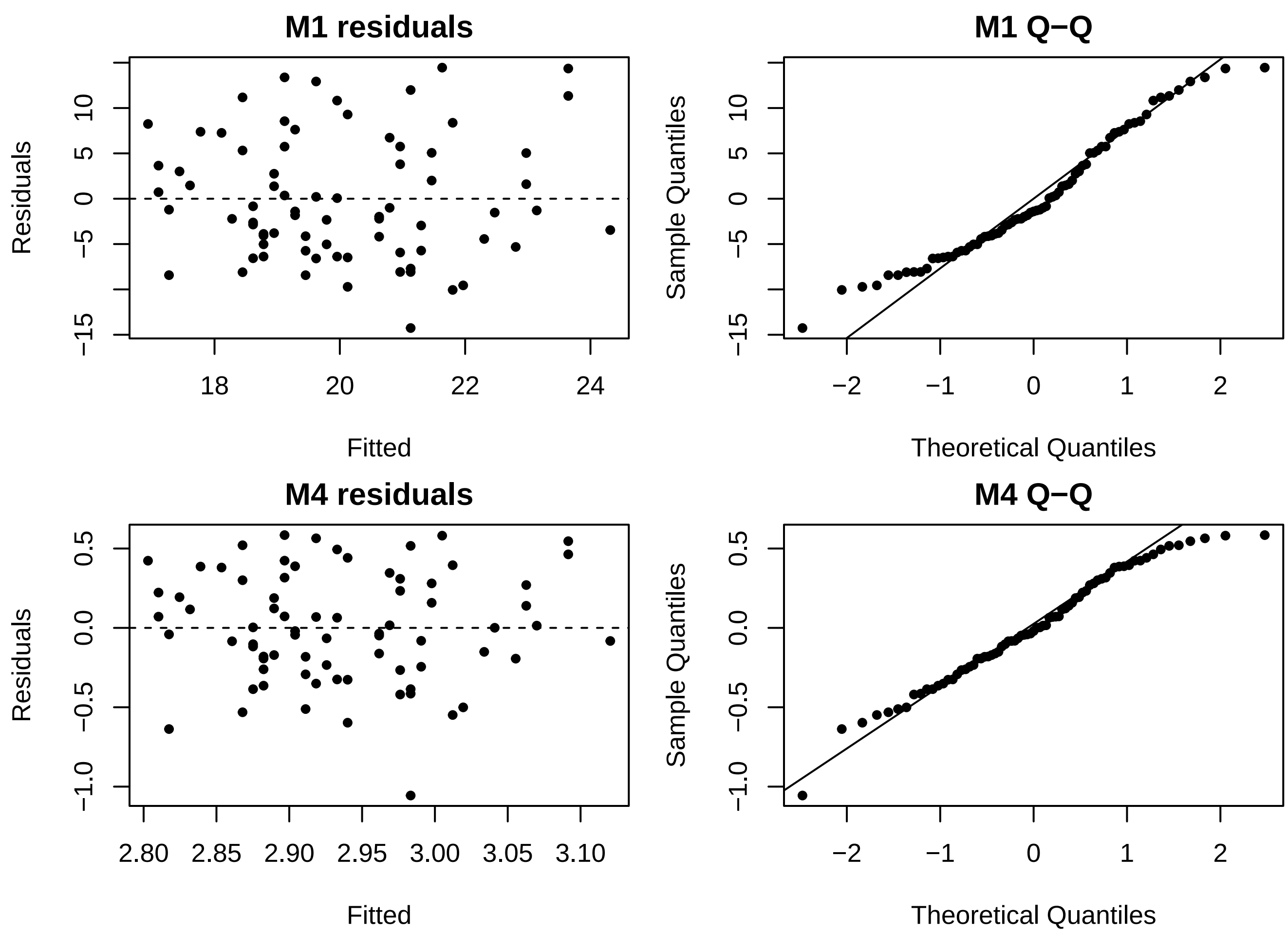

Figure 4: Diagnostic plots for Models 1 and 4.

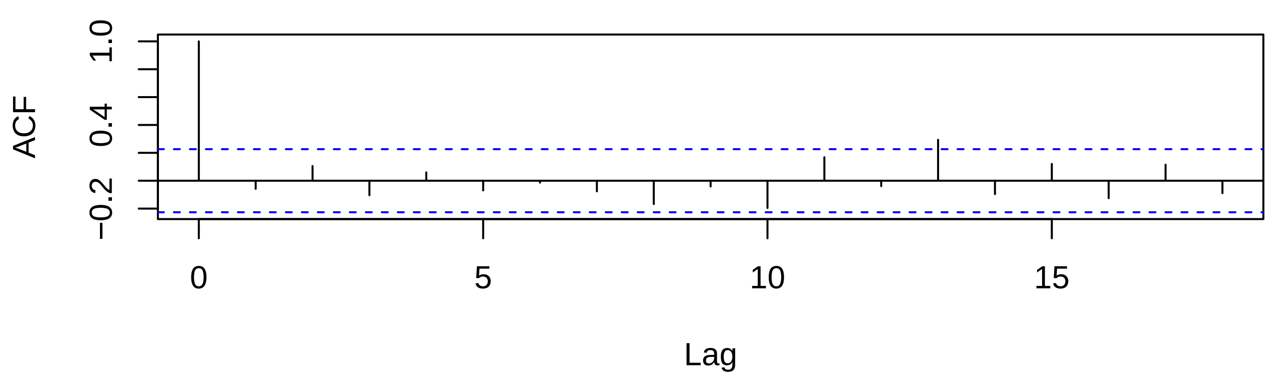

Figure 5: Residual autocorrelation, Model 1.